Idp5

Identification of the groups of order dividing p^5

1.0.0

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Abstract

The ldp5 package identifies the groups of order dividing p^5, for any prime p, with the groups in the SmallGroups library, and constructs isomorphisms onto the library groups.

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Chapter 1

Introduction

1.1 Overview

This package identifies the groups of order dividing p^5, for any prime p, with the groups in the SmallGroups library [BEO02], and it constructs isomorphisms onto the library groups. The algorithms are described in [DES].

The SmallGroups library contains all groups of order p^5, but its identification function `IdGroup` (**smallgrp: IdGroup**) is not available for the groups of order p^5 with $p \geq 7$. After loading ldp5, `IdSmallGroup` (**smallgrp: IdSmallGroup**) (equivalently `IdGroup` (**smallgrp: IdGroup**)) works for all groups of order p^n with $n \leq 5$; see Section 2.1. Moreover, the package provides the attribute `IsomorphismSmallGroup` (2.2.1), which constructs an isomorphism from a group G of order dividing p^5 onto `SmallGroup(IdSmallGroup(G))`.

The function `IdGroupsAvailable` (**smallgrp: IdGroupsAvailable**) is not changed by the package: for example, `IdGroupsAvailable(7^5)` still returns `false`.

The identification of a group of order dividing p^4 also yields an isomorphism onto the library group. The identification of a group of order p^5 uses invariants, the identification of quotients of order dividing p^4, and linear algebra for the families of groups that depend on parameters. An isomorphism from a group of order p^5 onto the library group is then constructed by lifting isomorphisms between quotients, partly with a randomised search; see Section 2.2.

The randomised lifting algorithm for the isomorphisms of groups of order p^5 has been developed with the LLM Claude Opus; every isomorphism that it returns is verified. The package files and this manual have been prepared with the assistance of Claude Opus.

1.2 An example

We identify a randomly generated permutation group isomorphic to `SmallGroup(7^5,65)`, and construct an isomorphism onto the library group.

Example

```
gap> R := SmallGroup(7^5, 65);;
gap> R := Image(IsomorphismPermGroup(R));;
gap> repeat gens := List([1..8], x -> Random(R)); until Group(gens) = R;
gap> R := Group(gens);;
gap> IdSmallGroup(R);
[ 16807, 65 ]
gap> iso := IsomorphismSmallGroup(R);;
```

```
gap> Source(iso) = R;
true
gap> IdSmallGroup(Range(iso));
[ 16807, 65 ]
```

Each call of `SmallGroup(7^5,65)` creates a new group, so `Range(iso)` is not equal to a group `H` constructed with `H := SmallGroup(7^5,65)`. However, both groups are given by the same pc presentation, so mapping `Pcgs(Range(iso))` onto `Pcgs(H)` defines an isomorphism.

Example

```
gap> H := SmallGroup(7^5, 65);;
gap> isoRtoH := iso * GroupHomomorphismByImages(Range(iso), H,
>                                  Pcgs(Range(iso)), Pcgs(H));;
gap> Source(isoRtoH) = R and Range(isoRtoH) = H and IsBijective(isoRtoH);
true
```

Chapter 2

Functions

2.1 Identification

The package installs a method for the attribute `IdSmallGroup` (**smallgrp: IdSmallGroup**) (a synonym of `IdGroup` (**smallgrp: IdGroup**)) of the SmallGroups library. For a group G of order p^n with $p > 5$ prime and $n \leq 5$, this method returns the pair `[p^n,i]` such that G is isomorphic to `SmallGroup(p^n,i)`. For all other groups, in particular for the groups of order 2^n, 3^n and 5^n, the methods of the SmallGroups library are used.

The input group can be given in any representation in which GAP can compute with it, for example as a pc group, a permutation group, or a matrix group. If an isomorphism onto the library group is found during the identification, which is the case for the groups of order dividing p^4, then it is stored as the value of `IsomorphismSmallGroup` (2.2.1).

Example

```
gap> G := Group(GeneratorsOfGroup(SmallGroup(11^5, 40)));;
gap> IdGroup(G);
[ 161051, 40 ]
gap> G := Image(RegularActionHomomorphism(SmallGroup(13^4, 9)));;
gap> IdSmallGroup(G);
[ 28561, 9 ]
gap> HasIsomorphismSmallGroup(G);
true
gap> IdGroupsAvailable(17^5);  # the SmallGroups library has no ID function
false
gap> NumberSmallGroups(17^5);
101
gap> ForAll([1..101], i -> IdGroup(Group(GeneratorsOfGroup(SmallGroup(17^5,i))))
>                          = [17^5, i]);
true
```

2.2 Isomorphisms onto the library groups

2.2.1 IsomorphismSmallGroup (for IsGroup)

▷ `IsomorphismSmallGroup(G)` (attribute)

Returns: an isomorphism from `G` onto `SmallGroup(IdSmallGroup(G))`, or `fail`

For a group `G` of order p^n with p prime and $n \leq 5$, this attribute is an isomorphism from `G` onto the library group `SmallGroup(IdSmallGroup(G))`.

- If $p > 5$ and $n \leq 4$, then the isomorphism is computed together with `IdSmallGroup(G)`.
- If $p \in \{2,3\}$, or if $p = 5$ and $n \leq 4$, then GAP's function `IsomorphismGroups` is used.
- If $n = 5$ and $p \geq 5$, then the isomorphism is constructed by lifting isomorphisms between quotients of `G` and of the library group, partly with a randomised search. The map that is returned is always verified to be an isomorphism. In principle the search can fail, and then `fail` is returned and a message is printed at level 1 of `InfoIdp5` (2.4.1); this has not been observed in practice. Since `IsomorphismSmallGroup` is an attribute, the value `fail` is stored; to try again, call `IsomorphismSmallGroup` for a new copy of the group, for example `Group(GeneratorsOfGroup(G))`.

Each call of `SmallGroup` creates a new group, so the range of the isomorphism is not equal to another group constructed with `SmallGroup(IdSmallGroup(G))`; but the two groups are given by the same pc presentation (see the example in Chapter 1).

Example

```
gap> G := Image(IsomorphismPermGroup(SmallGroup(7^5, 30)));;
gap> iso := IsomorphismSmallGroup(G);;
gap> IsBijective(iso) and Source(iso) = G;
true
gap> IdSmallGroup(Range(iso));
[ 16807, 30 ]
gap> G := DirectProduct(CyclicGroup(IsPermGroup, 4), DihedralGroup(IsPermGroup, 8));;
gap> iso := IsomorphismSmallGroup(G);;
gap> IdSmallGroup(Range(iso)) = IdSmallGroup(G);
true
```

2.3 Types of the groups of order p^5

For primes $p \geq 7$, the SmallGroups library constructs the groups of order p^5 from 70 parametrised pc presentations; the number $t \in \{1,\ldots,70\}$ of the presentation of a group is its *type*. The groups of a type form a consecutive range of IDs. Most types consist of a single group; the types 11, 12, 29, 32, 33, 48 and 50 have a further parameter k, and the pair $[t,k]$ determines a group of type t. The following two functions convert between the IDs and the types.

2.3.1 TypeFromIdSmallGroup_p5

▷ `TypeFromIdSmallGroup_p5(p, i)` (function)

Returns: a list `[t]` or `[t,k]`

For a prime $p > 5$ and an integer `i` in `[1..NumberSmallGroups(p^5)]`, this function returns the type of `SmallGroup(p^5,i)`: the list `[t]` if the type t has no further parameter, and the list `[t,k]` otherwise.

2.3.2 IdSmallGroupFromType_p5

▷ `IdSmallGroupFromType_p5(`*p, type*`)` (function)

Returns: an integer or a list of integers

This is the inverse of `TypeFromIdSmallGroup_p5` (2.3.1). For a prime $p > 5$, the argument *type* can be an integer t in `[1..70]` or a list `[t]`; then the IDs i of the groups `SmallGroup(`p`^5,i)` of type t are returned, as an integer if there is only one such group and as a list otherwise. If *type* is a pair `[t,k]`, then the ID of the k-th group of type t is returned.

Example

```
gap> TypeFromIdSmallGroup_p5(7, 34);
[ 29, 2 ]
gap> IdSmallGroupFromType_p5(7, [29, 2]);
34
gap> IdSmallGroupFromType_p5(7, 29);
[ 33, 34, 35 ]
gap> TypeFromIdSmallGroup_p5(7, 65);
[ 52 ]
gap> IdSmallGroupFromType_p5(7, 52);
65
gap> ForAll([1..NumberSmallGroups(7^5)],
>           i -> IdSmallGroupFromType_p5(7, TypeFromIdSmallGroup_p5(7, i)) = i);
true
```

2.4 The info class

2.4.1 InfoIdp5

▷ `InfoIdp5` (info class)

The info class of the package. At level 1 (the default), a message is printed if the randomised search of `IsomorphismSmallGroup` (2.2.1) fails. At levels 2 and 3, the construction of isomorphisms for groups of order p^5 prints which of its cases has been used, and some statistics.

2.5 Testing the package

The test suite of the package is run with `TestPackage("idp5")`. It identifies random copies of all groups of order p^n, $n \leq 5$, for several primes p, constructs isomorphisms onto the library groups, and runs the examples of this manual; this takes about a minute. The following function runs similar tests for given primes and prints runtimes.

2.5.1 TestSmallGroupID_p5

▷ `TestSmallGroupID_p5(`*lps, nr, testIso*`)` (function)

Returns: a record with the runtimes and the failures

For every prime p in the list *lps*, every e in `[1..5]` and every i in `[1..NumberSmallGroups(p^e)]`, this function constructs *nr* random isomorphic copies of `SmallGroup(p^e,i)` (pc groups with respect to a random pcgs), and checks that `IdSmallGroup` (**smallgrp: IdSmallGroup**) returns `[p^e,i]`. If *testIso* is `true`, then it also checks that

`IsomorphismSmallGroup` (2.2.1) returns an isomorphism onto the library group. For each copy, the runtimes in milliseconds are printed. For the groups of order p^5, the average runtimes and the ten largest runtimes are printed at the end, as well as the numbers of wrong IDs, of failed isomorphism searches, and of returned maps that are not isomorphisms. A wrong ID, and a map that is not an isomorphism, also cause an error.

The returned record has the components `id` and `iso` (lists of triples `[time, p, i]` for the groups `SmallGroup(p^5,i)`), `wrong`, `failed` and `notiso` (lists of the groups for which the ID was wrong, the isomorphism search failed, or the returned map was not an isomorphism). The runtimes depend on the machine.

Example

```
gap> t := TestSmallGroupID_p5([37, 61, 101], 1, true);;
------ start p = 37
ID [37^1,1] copy 1:   id 2 ms,  iso 0 ms
ID [37^2,1] copy 1:   id 1 ms,  iso 0 ms
ID [37^2,2] copy 1:   id 1 ms,  iso 0 ms
...
ID [101^5,267] copy 1:   id 1 ms,  iso 11 ms
ID [101^5,268] copy 1:   id 3 ms,  iso 31 ms
ID [101^5,269] copy 1:   id 1 ms,  iso 1 ms

------ summary
average time for ID: 51 ms   (607 tests)
ten largest runtimes for ID:
   273 ms   for SmallGroup(101^5,257)
   235 ms   for SmallGroup(101^5,256)
   ...
average time for iso: 71 ms   (607 tests)
ten largest runtimes for iso:
   3187 ms   for SmallGroup(101^5,125)
   1595 ms   for SmallGroup(101^5,119)
   ...
wrong IDs: 0
isomorphism search failed: 0
returned map not an isomorphism: 0
```

References

[BEO02] Hans Ulrich Besche, Bettina Eick, and E. A. O'Brien. A millennium project: constructing small groups. *Internat. J. Algebra Comput.*, 12:623–644, 2002. 4

[DES] Heiko Dietrich, Bettina Eick, and Henrik Schanze. Group identification for the groups of order dividing p^5. In preparation. 4

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